

Kubernetes Control Plane Troubleshooting Guide

1. API Server Down

- kubectl shows connection refused.
- Check kubelet: `systemctl status kubelet`
- Check API server: `crictl ps | grep kube-apiserver`
- Check logs: `journalctl -u kubelet -xe`
- Common causes: kubelet stopped, port 6443 blocked, expired cert.

2. etcd Failure

- Cluster becomes unresponsive.
- Check etcd: `crictl ps | grep etcd`
- Check logs: `crictl logs`
- Check disk: `df -h`
- Common causes: disk full, corruption, cert expired.

3. Scheduler Failure

- Pods stuck in Pending.
- Check scheduler container status.
- Review scheduler logs.

4. Controller Manager Failure

- ReplicaSets not creating pods.
- Check controller-manager logs.

5. Certificate Expired

- Error: x509 certificate expired.

- Run: `kubeadm certs check-expiration`
- Renew: `kubeadm certs renew all`
- Restart kubelet.

6. Disk Full

- Control plane components crash.
- Run: `df -h`
- Clean logs or increase disk.

7. Manifest Deleted

- Component disappears suddenly.
- Check: `/etc/kubernetes/manifests/`
- Restore from backup.

8. Firewall Issue

- Workers cannot reach API server.
- Ensure port 6443 open.

9. High Resource Usage

- Components restarting.
- Check with: `top`
- Look for OOMKilled events.

10. etcd Backup & Restore

- Backup: `ETCDCTL_API=3 etcdctl snapshot save backup.db`
- Always test restore in staging.
- Keep regular automated backups.